Mathematical Statistics

Statistical Decision

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Decision Rules

- Let X be a sample from a population $P \in \mathcal{P}$.
- A statistical decision is an action that we take after we observe X, for example, a conclusion about P or a characteristic of P.
- A be the set of allowable actions. Let $\mathcal{F}_A$ be a σ-field on A. Then the measurable space $(A, \mathcal{F})$ is called the action space.
- Let $\mathcal{X}$ be the range of X and $\mathcal{F}_{\mathcal{X}}$ be a σ-field on $\mathcal{X}$.
- A decision rule is a measurable function (a statistic) T from $(\mathcal{X}, \mathcal{F}_X)$ to $(A, \mathcal{F}_A)$.
- If a decision rule T is chosen, then we take the action $T(X) \in A$ whenever X is observed.

Decision Rules

- The construction or selection of decision rules cannot be done without any criterion about the performance of decision rules.
- Statistical decision theory sets a criterion using a loss function L as a function from $\mathcal{P} \times A$ to $[0, \infty)$ and is Borel on $(A, \mathcal{F}_A)$ for each fixed $P \in \mathcal{P}$.

Loss Functions for One Parameter

- Let $L(\theta, \widehat{\theta})$ be a loss function, representing how good the estimator is.
- Some common loss functions for one-dimensional parameter:

$L(\theta, \widehat{\theta}) = (\theta - \widehat{\theta})^2$	squared error loss
$L(\theta, \widehat{\theta}) = \lvert\theta - \widehat{\theta}\rvert^p$	L_p loss
$L(\theta, \widehat{\theta}) = 0$ if $\theta = \widehat{\theta}$ or 1 if $\theta \neq \widehat{\theta}$	zero-one loss
$L(\theta, \widehat{\theta}) = \int \log(\frac{p(x;\theta)}{p(x;\widehat{\theta})}) p(x;\theta)\, dx$	Kullback-Leibler loss

Loss Functions for Parameter Vectors

If $\theta = (\theta_1, \ldots, \theta_K)$ is a vector, some common loss functions:

- $$L(\theta, \widehat{\theta}) = \|\theta - \widehat{\theta}\|^2 = \sum_{j=1}^{K} (\widehat{\theta} - \theta_j)^2.$$
- $$L(\theta, \widehat{\theta}) = \|\theta - \widehat{\theta}\|_p = \left(\sum_{j=1}^{K} (\widehat{\theta} - \theta_j)^p \right)^{1/p}.$$

Example Rules

- (Estimation) We like to make a decision on the value of $\theta \in \Omega$, based on the sample $X = (X_1, \ldots, X_n)$. An example of a decision rule is $T(X) = \bar{X}$, the sample mean.
- A common loss function in this problem is the squared error loss $L(\theta, a) = (\theta - a)^2$, $a \in A$.
- (Hypothesis Testing): $H_0 : P \in \mathcal{P}_0$ versus $H_1 : P \in \mathcal{P}_1$.
- The action space $A = \{0, 1\}$. The decision rule $T(X)$ is a mapping from $\mathcal{X}$ to $\{0, 1\}$ and must have the form $I_C(X)$, where $C \in \mathcal{F}_{\mathcal{X}}$ is the rejection region.

Another Decision Rule Example

- A hazardous toxic waste site requires clean-up when the true chemical concentration θ in the contaminated soil is higher than a given level $\theta_0 \geq 0$.
- Because of the limitation in resources, we would like to spend our money and efforts more in those areas that pose high risk to public health.
- In a particular area where soil samples are obtained, we would like to take one of these three actions: a complete clean-up (a_1), a partial clean-up (a_2), and no clean-up (a_3). Thus, $A = \{a_1, a_2, a_3\}$.
- Suppose that the cost for a complete clean-up is c_1 and for a partial clean-up is $c_2 < c_1$; the risk to public health is $c_3(\theta - \theta_0)$ if $\theta > \theta_0$ and 0 if $\theta \leq \theta_0$.

Another Decision Rule Example

- Complete clean-up can reduce the toxic concentration to an amount $\leq \theta_0$, a partial clean-up can only reduce a fixed amount of the toxic concentration, i.e., the chemical concentration becomes $\theta - t$ after a partial clean-up, where t is a known constant.
- Loss function:

$L(\theta, a)$	a_1	a_2	a_3
$\theta \leq \theta_0$	c_1	c_2	0
$\theta_0 < \theta \leq \theta_0 + t$	c_1	c_2	$c_3(\theta - \theta_0)$
$\theta > \theta_0 + t$	c_1	$c_2 + c_3(\theta - \theta_0 - t)$	$c_3(\theta - \theta_0)$

Randomized Decision Rules

- A randomized decision rule is a function δ on $\mathcal{X} \times \mathcal{F}_A$ such that, for every $A \in \mathcal{A}$, $\delta(\cdot, A)$ is a Borel function and for every $x \in \mathcal{X}$, $\delta(x, \cdot)$ is a probability measure on $(A, \mathcal{A})$.

Risk Function

- Let $X_1, \ldots, X_n \sim P_\theta(.)$, where $\theta \in \Omega$
- The risk (function) of an estimator $\widehat{\theta}$ is the expected loss:

$$R(\theta, \widehat{\theta}) = E_\theta(L(\theta, \widehat{\theta}))$$

$$= \int L(\theta, \widehat{\theta}(x_1, \ldots, x_n)) p(x_1, \ldots, x_n; \theta)\, dx,$$

  which is a function of θ.

- When the loss function is squared error, the risk is just the MSE (mean squared error):

$$R(\theta, \widehat{\theta}) = E_\theta(\widehat{\theta} - \theta)^2 = Var_\theta(\widehat{\theta}) + Bias^2.$$

Optimal Rules

- A rule $\widehat{\theta}_1$ is as good as another rule $\widehat{\theta}_2$ if and only if

$$R(\theta, \widehat{\theta}_1) \leq R(\theta, \widehat{\theta}_2), \theta \in \Omega$$

- A rule $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are equivalent if and only if

$$R(\theta, \widehat{\theta}_1) = R(\theta, \widehat{\theta}_2), \theta \in \Omega.$$

- Let $\mathcal{T}$ be a class of allowable decision rules. Then, T^* is $\mathcal{T}$-optimal if T^* is as good as any other rule in $\mathcal{T}$.

Inadmissibility

Consider a given decision problem with a given loss $L(\theta, a)$.

- Inadmissibility Definition: An decision rule δ is inadmissible if there exists another rule $\delta^{'}$ such that

$$R(\theta, \delta') \leq R(\theta, \delta) \text{ for all } \theta, R(\theta, \delta') < R(\theta, \delta) \text{ for at least one } \theta,$$

- Admissible rule is not inadmissible rule.

Comparison of Two Rules and Admissible Rule

- To compare two estimators, we compare their risk functions. However, this does not provide a clear answer as to which estimator is better.
- Let $X \sim N(\theta, 1)$ and assume that we are using squared error loss.
- We consider two estimators: $\hat{\theta}_1 = X$ and $\hat{\theta}_2 = 3$. The risk functions are $R(\theta, \hat{\theta}_1) = 1$ and $R(\theta, \hat{\theta}_2) = (3-\theta)^2$. Therefore, $R(\theta, \widehat{\theta}_2) < R(\theta, \widehat{\theta}_1)$ if $2 < \theta < 4$, and $R(\theta, \widehat{\theta}_1) < R(\theta, \widehat{\theta}_2)$ otherwise.
- Furthermore, we will show that $\widehat{\theta}_2$ is admissible. If it were not, there exists a different rule δ' with smaller risk. In particular, $R(3, \delta') \leq R(3, \widehat{\theta}_2) = 0$. Hence, $0 = \int (\delta'(x) - 3)^2 f(x; 3)\, dx$. Thus, $\delta'(x) = 3$.

Risk Functions

- This example highlights the need to be able to compare risk functions. To do so, we need a one-number summary of the risk function.
- Two such summaries are the maximum risk and the Bayes risk.

Inadmissibility and Optimality

Consider a given decision problem with a given loss $L(\theta, a)$. Rao-Blackwell Theorem:

- (Inadmissibility) Let T be a sufficient statistic for $\theta \in \Omega$, $T_0 \in \mathcal{R}^k$, a non-randomized rule satisfying $E\| T_0\| < \infty$ and $T_1 = E(T_0(X) \mid T)$. Then, $R(\theta, T_1) \leq R(\theta, T_0)$ for any $\theta \in \Omega$. If $L(\theta, a)$ is strictly convex in a and T_0 is not a function of T, then T_0 is inadmissible.
- The proof of Theorem is an application of Jensen's inequality.

Inadmissibility and Optimality

- The concept of admissibility helps us to eliminate some decision rules. However, usually there are still too many rules left after the elimination of some rules according to admissibility and sufficiency.
- Since an optimal rule often does not exist, we may use the following two approaches to choose a decision rule.
- The first approach is to define a class $\mathcal{T}$ of decision rules that have some desirable properties (statistical and/or nonstatistical) (e.g. unbiasedness) and then try to find the best rule in $\mathcal{T}$.
- The second approach to finding a good decision rule is to consider some characteristic $R(T)$ of $R(P, T)$, for a given decision rule T, and then minimize $R(T)$ over $T \in \mathcal{T}$: (1) Bayes rule, and (2) minimax rule.

Maximum and Bayes Risk

- The maximum risk:

$$\bar{R}(\widehat{\theta}) = \sup_{\theta \in \Theta} R(\theta, \widehat{\theta}).$$

- The Bayes risk under prior π:

$$B_\pi(\widehat{\theta}) = \int R(\theta, \widehat{\theta}) \pi(\theta) \, d\theta.$$

Example

- Let $X_1, \ldots, X_n \sim Bernoulli(p)$.
- Consider two estimators, $\widehat{p}_1 = \bar{X}$, and $\widehat{p}_2 = \frac{Y+\alpha}{\alpha+\beta+n}$, where $Y = \sum_{i=1}^n X_i$ and $\alpha = \beta = \sqrt{n/4}$.
- $$\bar{R}(\widehat{p}_1) = \max_{0 \le p \le 1} \frac{p(1-p)}{n} = \frac{1}{4n},$$
$$\bar{R}(\widehat{p}_2) = \max_{0 \le p \le 1} \frac{n}{4(n+\sqrt{n})^2} = \frac{n}{4(n+\sqrt{n})^2}.$$
- Hence, $\widehat{p}_2$ is a better estimator.
- However, when n is large, $R(p, \widehat{p}_1)$ has smaller risk except for a small region in the parameter space near $p = 1/2$. Thus, many people prefer $\widehat{p}_1$ to $\widehat{p}_2$.

Bayes Estimator

- An estimator θ that minimizes the Bayes risk is called a Bayes estimator. That is

$$B_\pi(\widehat{\theta}) = \inf_{\tilde{\theta}} B_\pi(\tilde{\theta})$$

where the infimum is over all estimators $\tilde{\theta}$.

Minimax Estimator

- An estimator that minimizes the maximum risk is called a minimax estimator. That is,

$$\sup_{\theta} R(\theta, \widehat{\theta}) = \inf_{\tilde{\theta}} \sup_{\theta} R(\theta, \tilde{\theta}),$$

  where the infimum is over all estimators $\tilde{\theta}$.

- We call the right hand of the above equation equation,

$$\sup_{\theta} R(\theta, \widehat{\theta}) = \inf_{\tilde{\theta}} \sup_{\theta} R(\theta, \tilde{\theta}),$$

  the maximum risk.

Decision Theory

- Statistical decision theory: (1) determine the minimax risk $\bar{R}$ and (2) find an estimator that achieves this risk.
- After we have found the minimax risk, we want to find the minimax estimator that achieves this risk:

$$\sup_{\theta\in\Theta} R(\theta,\widehat{\theta}) = \inf_{\tilde{\theta}} \sup_{\theta\in\Theta} R(\theta,\tilde{\theta}).$$

- We may find a minimax estimator asymptotically,

$$\sup_{\theta\in\Theta} R(\theta,\widehat{\theta}_n) \sim \inf_{\tilde{\theta}} \sup_{\theta\in\Theta} R(\theta,\tilde{\theta}), \text{ as n} \to \infty,$$

where $a_n \sim b_n$ means that $a_n/b_n \to 1$.

- Or, we may find an estimator that can achieve the minimax rate:

$$\sup_{\theta\in\Theta} R(\theta,\widehat{\theta}) \asymp \inf_{\tilde{\theta}} \sup_{\theta\in\Theta} R(\theta,\tilde{\theta}), \text{ as n} \to \infty,$$

where $a_n \asymp b_n$ mean that a_n/b_n and b_n/a_n are both bounded.

How to Find Bayes Estimators

- Let π be a prior distribution.After observing $x^n = (x_1, \ldots, x_n)$, the posterior distribution is,

$$P(\theta \in A \mid x^n) = \frac{\int_A p(x_1, \ldots, x_n \mid \theta)\pi(\theta)\, d\theta}{\int p(x_1, \ldots, x_n \mid \theta)\pi(\theta)\, d\theta} = \frac{\int_A L(\theta)\pi(\theta)\, d\theta}{\int_\Theta L(\theta)\pi(\theta)\, d\theta},$$

  where $L(\theta) = p(x^n; \theta)$ is the likelihood.

- The posterior has density

$$\pi(\theta \mid x^n) = \frac{p(x^n \mid \theta)\pi(\theta)}{m(x^n)},$$

  where $m(x^n) = \int p(x^n \mid \theta)\pi(\theta)\, d\theta$ is the marginal distribution of X^n.

How to Find Bayes Estimators

- The posterior risk of an estimator $\widehat{\theta}(x^n)$:

$$r(\widehat{\theta} \mid x^n) = \int L(\theta, \widehat{\theta}) \pi(\theta \mid x^n) \, d\theta.$$

- Theorem: For each x^n, $\widehat{\theta}(x^n)$ is the value of θ that minimizes $r(\widehat{\theta} \mid x^n)$. Then $\widehat{\theta}(x^n)$ is the Bayes estimator.

Proof

Note that Bayes risk can be rewritten as follows:

$$
\begin{aligned}
B_n(\widehat{\theta}) &= \int R(\theta, \widehat{\theta}) \pi(\theta)\, d\theta \\
&= \int \left(\int L(\theta, \widehat{\theta}(x^n)) p(x^n \mid \theta)\, d\theta \right) \pi(\theta)\, d\theta \\
&= \int \int L(\theta, \widehat{\theta}(x^n)) \pi(\theta \mid x^n) m(x^n)\, dx^n\, d\theta \\
&= \int \left(\int L(\theta, \widehat{\theta}(x^n)) \pi(\theta \mid x^n)\, d\theta \right) m(x^n)\, dx^n \\
&= \int r(\widehat{\theta} \mid x^n) m(x^n)\, dx^n.
\end{aligned}
$$

So, if we choose $\widehat{\theta}(x^n)$ to be the value of θ that minimizes $r(\widehat{\theta} \mid x^n)$, then we will also minimize the integral $\int r(\widehat{\theta} \mid x^n) m(x^n)\, dx^n$.

Bayes Estimators for Some Special Loss Functions

- If $L(\theta, \widehat{\theta}) = (\theta - \widehat{\theta})^2$, the Bayes estimator is

$$\widehat{\theta}(x^n) = \int \theta \pi(\theta \mid x^n) \, d\theta = E(\theta \mid X = x^n).$$

- If $L(\theta, \widehat{\theta}) = \mid \theta - \widehat{\theta} \mid$, the Bayes estimator is the median of the posterior $\pi(\theta \mid x^n)$.
- If $L(\theta, \widehat{\theta})$ is zero-one loss, then the Bayes estimator is the mode of the posterior $\pi(\theta \mid x^n)$.

Proof

Proof for the squared error loss. Note that the Bayes estimator $\widehat{\theta}(x^n)$ can be found by minimizing

$$r(\delta \mid x^n) = \int (\delta - \theta)^2 \pi(\theta \mid x^n)\, d\theta$$

Taking the derive of $r(\delta \mid x^n)$ with respect to δ and setting it to zero yields the equation

$$2 \int (\theta - \delta)^2 \pi(\theta \mid x^n) = 0.$$

Solving this equation for δ leads to $\delta = E(\theta \mid X^n = x^n)$.

Homework for proving the other two types of loss functions.

Example

- Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where σ^2 is known.
- Suppose μ has a prior distribution $N(a, b^2)$.
- Then, the Bayes estimator with respect to squared error loss is the posterior mean, which is

$$\widehat{\theta}(X_1, \ldots, X_n) = \frac{b^2}{b^2 + \sigma^2/n} \bar{X} + \frac{\sigma^2/n}{b^2 + \sigma^2/n}.$$

Connection between Bayes Estimator and Minimax Estimator

- Finding minimax estimators is complicated.
- But, Bayes estimators with a constant risk function are minimax.
- Theorem: Suppose that $\widehat{\theta}$ is the Bayes estimator with respect to some prior. If risk is constant, then $\widehat{\theta}$ is a minimax.

Connection between Bayes Estimator and Minimax Estimator

- Lemma: Let $\widehat{\theta}$ be the Bayes estimator for some prior π. If $R(\theta, \widehat{\theta}) \leq B_\pi(\widehat{\theta})$ for all θ, then $\widehat{\theta}$ is minimax and π is called a least favorable prior.
- Proof: Suppose $\widehat{\theta}$ is not minimax. Then, there is another estimator $\widehat{\theta}_0$ such that $\sup_\theta R(\theta, \widehat{\theta}_0) < \sup_\theta R(\theta, \widehat{\theta})$. Since $B_\pi(\widehat{\theta}_0) = \int R(\theta, \widehat{\theta}_0)\pi(\theta)\, d\theta$,

$$B_\pi(\widehat{\theta}_0) \leq \sup_\theta R(\theta, \widehat{\theta}_0) < \sup_\theta R(\theta, \widehat{\theta}),$$

  which implies $B_\pi(\widehat{\theta}_0) < B_\pi(\widehat{\theta})$, a contradiction occurs.
- This concludes the proof of the lemma. Using the result of this lemma, we can show the theorem.

Example

- Consider the Bernoulli $B(n, p)$ with loss function

$$L(p, \widehat{p}) = \frac{(p - \widehat{p})^2}{p(1-p)}.$$

- Let $\widehat{p}(X^n) = \widehat{p} = \sum_{i=1}^n X_i/n$. The risk is

$$R(p, \widehat{p}) = E\left(\frac{(\widehat{p} - p)^2}{p(1-p)}\right) = \frac{1}{n},$$

  which, as a function of p, is constant. We can also show for this loss function, $\widehat{p}(X^n)$ is the Bayes estimator under the prior $\pi(p) = 1$.

- Hence, $\widehat{p}$ is minimax.

Maximum Likelihood Estimator

- For parametric models that satisfy weak regularity conditions, the maximum likelihood estimator is approximately minimax.
- Consider squared error loss which is squared bias plus variance. In parametric models with large samples, it can be shown that the variance term dominates the bias (typically, the squared bias is order $O(n^{-2})$ and the variance is of order $O(n^{-1})$ so the risk of the mle $\widehat{\theta}$ roughly equals the variance

$$R(\theta, \widehat{\theta}) = Var_\theta(\widehat{\theta}) + Bias^2 \approx Var_\theta(\widehat{\theta}).$$

- The variance of the MLE is approximately $Var(\widehat{\theta}) \approx \frac{1}{nI(\theta)}$, where $I(\theta)$ is the Fisher information.
- Therefore,

$$nR(\theta, \widehat{\theta}) \approx \frac{1}{I(\theta)}.$$

Maximum Likelihood Estimator, continued

- For any other estimator θ^*, it can be shown that for large n, $R(\theta, \theta^*) \geq R(\theta, \widehat{\theta})$.
- So the maximum likelihood estimator is approximately minimax. This assumes that the dimension of p is fixed and n is increasing.

An Example on the Hodges Estimator

- We give an example on the subtleties of optimal estimators.
- Let $X_1, \ldots, X_n \sim N(\theta, 1)$. Then, the MLE is $\widehat{\theta} = \bar{X}_n$.
- Let us consider another estimator due to Hodges. Let $J_n = [-\frac{1}{n^{1/4}}, \frac{1}{n^{1/4}}]$, and define the following estimator:

$$\tilde{\theta}_n = \begin{cases} \bar{X}_n & \text{if } \bar{X}_n \text{ does not in } J_n, p \\ 0 & \text{if } \bar{X}_n \in J_n. \end{cases}$$

- If $\theta \neq 0$, we choose ϵ so small so that 0 is not contained in $I = (\theta - \epsilon, \theta + \epsilon)$. By the law of large numbers, we have $P(\bar{X}_n \in I) \to 1$.

An Example on the Hodges Estimator, continued

- Since J_n is shrinking, for large n, $\tilde{\theta}_n = \bar{X}$ with high probability. We conclude that, for any $\theta \neq 0$, $\tilde{X}_n$ behaves like X_n.
- When $\theta = 0$,

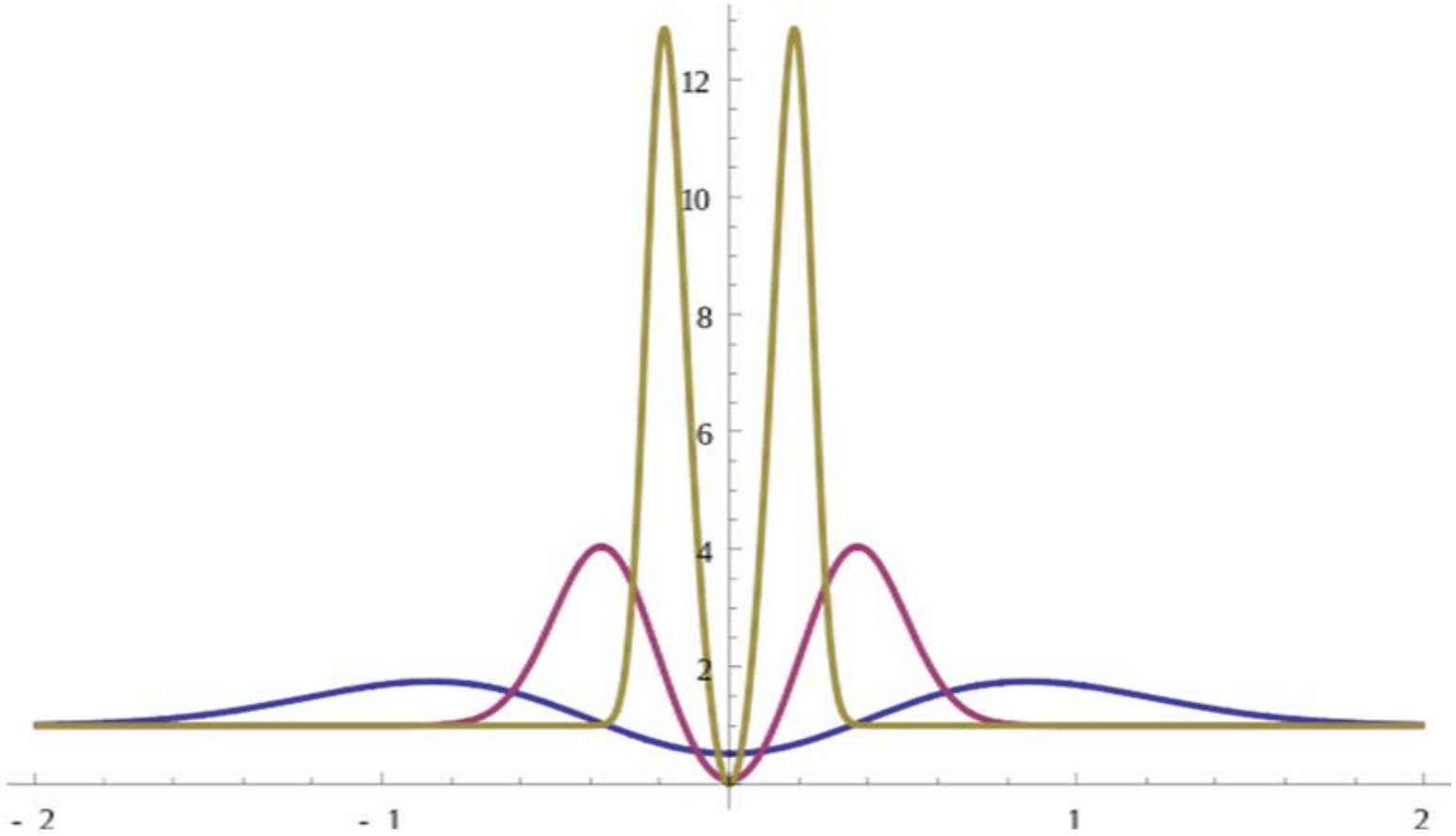
$$P(\bar{X}_n \in J_n) = P(\mid \bar{X}_n \mid \leq n^{-1/4})$$

$$= P(\sqrt{n} \mid \bar{X}_n \mid \leq n^{1/4}) = P(\mid N(0,1) \mid \leq n^{1/4}) \to 1.$$

- Hence, for large n, $\tilde{\theta} = \theta = 0$ with high probability.
- We conclude that Hodges estimator is like $\bar{X}_n$ when $\theta \neq 0$ and is better than $\bar{X}_n$ when $\theta = 0$. So, $\bar{X}_n$ is not the best estimator.
- Is it true?

Risk of the MLE and Hodge Estimator

- Figure below shows the nMSE or risk, $R_n(\theta) = E(\tilde{\theta}_n - \theta)^2$ as a function of θ.



- nMSE of the Hodges estimator for $n = 5$ (blue), $n = 50$ (purple), and $n = 500$ (olive). Note that $n\text{MSE}(\bar{X}_n) = 1$.

Lessons Learned

- The risk of $\tilde{\theta}$ is excellent at $\theta = 0$.
- At any other θ, it will eventually behave like the risk of $\bar{X}_n$.
- But the maximum risk of $\tilde{\theta}$ is terrible. We pay for the improvement at $\theta = 0$ by an increase in risk elsewhere.

Lessons Learned

- There are two lessons here.
- First, we need to pay attention to the maximum risk.
- Second, it is better to look at uniform asymptotics $\lim_{n\to\infty}\sup_\theta R_n(\theta)$ than pointwise asymptotics $\sup_\theta \lim_{n\to\infty} R_n(\theta)$.